

equipment. The software layer will act as a communications interface between the two, reducing the dependence of proprietary interfaces to integrate networks and test equipment.

Probes: Sharper than an eagle's eye

Probes are becoming ultra-fine, super-fast and 3-D-capable. Consider some of the recent features being touted by popular brands. There are double-sided, 16-probe flying systems with 3-D probing and analogue, digital and boundary scan test capability on all probes. Then, there are systems for ultra-fine-pitch probing of loaded PCBs that can test down to 24mil centres and 15mil diameter targets, enabling 90 per cent better test coverage on dense, loaded PCBs.

Software has also begun to play an increased role in this space, enabling so-called 'probe analysis.' Such systems basically add an interactive, rules-based routine to assist layout engineers and test developers in identifying usable PCB locations for physical access. These also generate a comprehensive accessibility report that states whether adequate probe locations were identified for each net, highlighting mechanical rules violation, so that this information can be used for design improvements.

Phased-array ultrasonics: A nascent technology worth watching

According to a recent report by Frost & Sullivan, phased array will be the future of ultrasonic non-destructive testing. Although this technology has been spoken about for quite some time now, it is likely to gain traction in the near future and is therefore worth a watch.

A phased array is an advanced method of executing ultrasonic scans. Instead of a single transducer element, phased-array systems use an array of elements that can be pulsed sequentially. By creating interfering wave patterns, it is possible to electronically manipulate beams so that their direction and angle can be controlled. This enables more

Today's 3G and 4G handsets and other portable devices include multiple radios representing numerous standards. Traditional test methods involve attaching dummy loads to the handsets and testing the various radios sequentially. New systems, on the other hand, will test all these in parallel

precise software control, inspection from multiple angles and inspection of complex geometries. Thanks to these abilities, experts are betting on the phased array to be the next big thing in non-destructive testing.

Handheld instrumentation: From the lab to the field

While handheld instrumentation has been around for a few years now, the last year has seen their increased application, especially in wireless and pharmaceutical industries.

Consider the use of near-infrared spectrometers for testing certain chemicals in the pharmaceuticals industry. The removal of cartons to labs and their repacking used to cause several disturbances including introduction of dust between the satchels. However, the availability of handheld instruments has now made 100 per cent inline testing possible today, avoiding a lot of unnecessary overhead.

Similarly, a range of wireless, handheld instruments have made it extremely easy to test the various components of next-generation networks.

Embedded instrumentation: Micro-instruments, macro benefits

Consider T&M instruments embedded as an additional feature in a chip. Embedded instruments are being recognised as a more cost-effective, efficient and agile solution to T&M needs of computing and communications industries.

As semiconductor devices and boards become more complex and crowded, the efficiency of external equipment and physical probes reduces, creating the need for embedded instrumentation. Plus, external equipment also cause certain issues in some situations.

For example, placing the probe of an oscilloscope on a test pad on a high-speed serial bus could introduce capacitance anomalies on the bus. It would become difficult for the engineer to identify whether the anomaly was caused by the instrument or a design fault.

Intel's Interconnect Built In Self Test (IBIST) is a good example of embedded instrumentation technology. Apart from this, companies such as Synopsis, Xilinx, Altera and Maxim also offer embedded instrumentation intellectual properties.

The Institute of Electrical and Electronics Engineers is now working on the Internal Joint Test Action Group (IJTAG) standard for embedded instrumentation. This standard will develop a methodology for access to embedded test and debug features, via the IEEE 1149.1 Test Access Port (TAP) and additional signals that may be required. The elements of the methodology would include a description language for characteristics of the features and for communication with the features, and requirements for interfacing to the features.

There is a lot more happening in video tests, virtual instrumentation, radar testing and beyond. Being an industry that serves numerous industries ranging from information and communication technology to aviation, aerospace and defence, in forms ranging from bench and modular instrumentation to machine vision and automated inspection, in scales ranging from component testing to complete system monitoring, it is obvious that there is scope for innovation every day—and trust me, it is happening at that pace! ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Bengaluru